

Site Scanning Instructions for Use of the MR Phantom for the ACR[™] MRI Accreditation Program

12/02



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ACR MRI Accreditation Program Phantom Version 2

Please note: The image on the other side of this page, is a sample image of version 2 of the ACR MRI Accreditation Program Phantom. If you have purchased a phantom, and have received version 2 with serial numbers: 2001-2113, and 2401 and higher, your images may look similar to those in the image sample on the other side of this page. If your site purchased version 1 of the ACR phantom, then please disregard this insert, and you may use your current phantom for accreditation testing, or for quality control purposes. For sites that have version 2, you may notice some minor differences in the images you obtain of your phantom in comparison to those in the site scanning instruction booklet. See the arrows and corresponding captions on the sample image for information regarding this issue. Although there are slight differences between version 1 and 2, there is no change in the actual dimensions of the phantom analyzed during the image review process by the ACR. Both versions can be used for accreditation testing and quality control. Sites can still use all the documents, and information contained in both the Site Scanning Instruction Booklet, as well as the Phantom Test Guidance Booklet enclosed in the testing package to prepare their image submission.

If you have any questions about the information above, please contact the MRI Accreditation department at (800) 770-0145.

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INTRODUCTION

The intent of the MRI Accreditation Program is to use the information obtained from the review of both clinical and phantom images to assess overall image quality. Your facility will need to perform two specified phantom scans using ACR protocols as well as two phantom scans using your site's routine clinical head protocol as outlined in Section 2 of this instruction book.

It has come to our attention that some manufacturer's of MRI systems have sent "sample" or "recommended" phantom site scanning protocols to their users. *Please be aware that the requirement for MRI accreditation is that in the second set of scans mentioned above, facilities use the same protocol for the phantom that the facility uses for head imaging. Failure to comply with this requirement could result in failure to achieve accreditation.*

Please review the following site scanning instructions and follow them carefully before obtaining images for submission to the accreditation program. If you have any questions about the site scanning instructions please contact the ACR.

1. PHANTOM SET-UP AND ALIGNMENT FOR SCANNING

The MRI Accreditation Phantom should be scanned in the head coil with the cylindrical phantom aligned as a head would be in the coil. Transaxial slices should result in circular crosssections of the phantom. The phantom should be positioned so that the word "Nose" is where the nose would be for a standard head study and the word "Chin" is where the chin would be located in a standard head study. The center of the phantom (the dark notch on the side of the phantom) should be placed in the center of the head coil and aligned with the positioning indicator light so that it will be in the isocenter of the scanner. Once grossly positioned, it is then necessary to "fine tune" the position of the phantom along all three axes. For this, you will need to use the non-metallic bubble level enclosed. Place the level along the top of the phantom running in and out of the scanner (along the z-axis) to ensure that the phantom is horizontal. Place a gauze pad under either end of the phantom to level the phantom horizontally. Next,

place the level on top of the plastic bar at the chin surface, rotating the phantom so that the plastic bar is horizontal. With the phantom then clamped or wedged inside the head coil, check to see that the sagittal laser alignment light is parallel to the line running along the "nose" surface of the phantom. (To see the laser light reflection, it may be necessary to place a piece of white paper on top of the phantom.) After each position adjustment, recheck that the top of the phantom and the chin bar are still horizontal. After the phantom has been moved into the center of the magnet, verify its positioning by performing sagittal and, if desired, coronal plane localizer scans, until correct. (Please note, some systems require that a weight be entered in order to scan the phantom. The ACR recommends that your site enter a weight of 200 lbs.) Once correctly aligned, the phantom should be kept in the same position during the entire series of scans.

2. SCANNING THE PHANTOM

A sagittal locator sequence should be acquired with the acquisition parameters listed on the Site Scanning Data Form. Use exactly these pulse sequence parameters, if possible, placing a check mark under each prescribed parameter to indicate that it has been used. If alternative parameters must be used because of machine or software limitations, enter the alternative scan parameters actually used below the ACR prescribed scan parameters. Fill in alternative parameters only for those parameters that differ

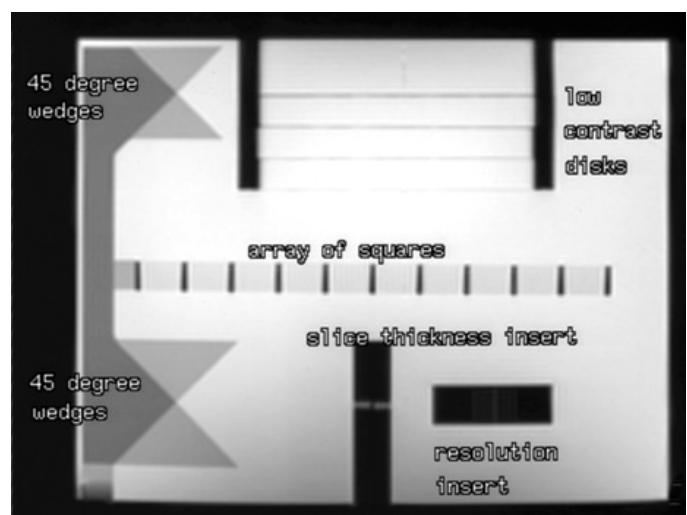


Figure 1: Sagittal localizer view of ACR MRI Phantom with several inclusions of the phantom labeled.

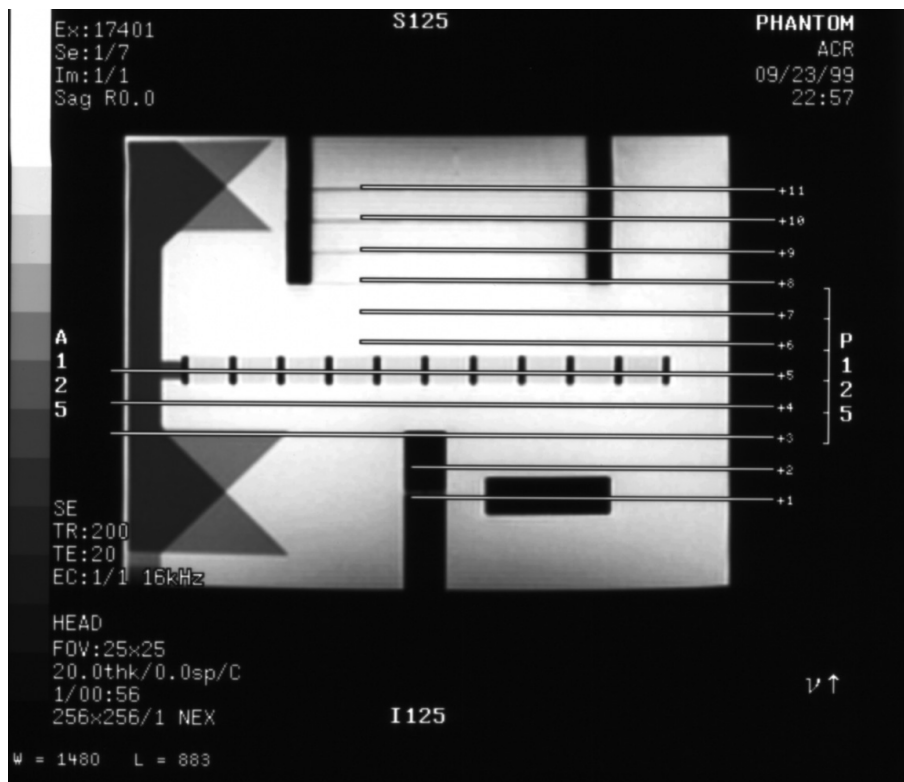


Figure 2: Sagittal locator image with slice locations for transaxial scans indicated.

from the ACR prescribed parameters. Deviations from the specified imaging parameters will often require a different overall study time. List the actual scan time required on the data form.

The sagittal locator scan should result in an image similar to Figure 1. If the pairs of 45° crossed wedges are not visible in the scan, the phantom must be repositioned and rescanned. A horizontal line used for slice prescription (see Figure 2) should be parallel to the low contrast disks located at the top of Figure 1 or Figure 2. If not, the phantom must be repositioned.

The next two scan acquisitions are transaxial pulse sequences acquired with identical spatial parameters: 5 mm slice thickness, 5 mm gap, 25 cm FOV, 256 x 256 matrix. At least 11 slices should be obtained, aligned using graphic prescription from the sagittal locator as shown in Figure 2 (Note: This is the preferred method for slice positioning). The center of slice #1 should be aligned with the vertex of the crossed wedges (visible on the lower left in Figures 1 and 2) and through the center of the dark chemical shift and resolution insert (visible on the lower right). Slice #1 should result in a transaxial image that looks like Figure 3. The centers of slices #8–11 should align with the four low-contrast discs shown toward the top in Figures 1 and 2. (Record this sagittal locator image,

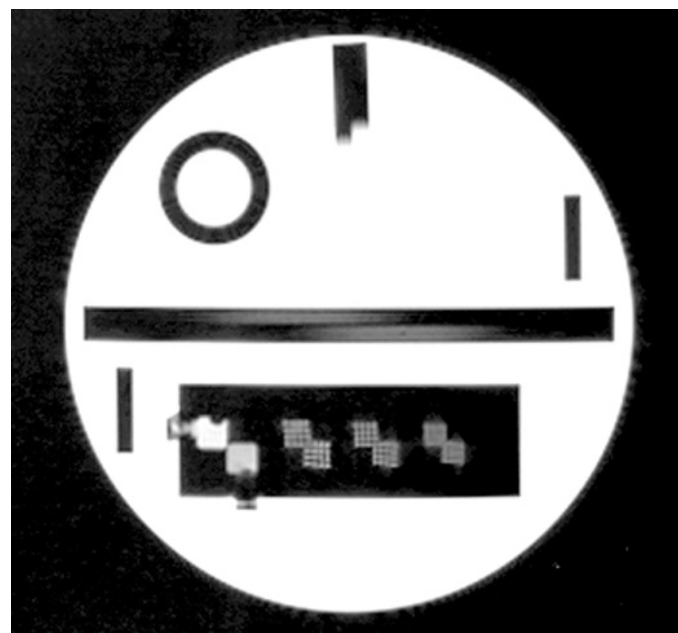


Figure 3: Slice #1 of the SE 500/20 transaxial scan.

using a 12 on 1 format, showing the locations of the prescribed transaxial slices.) If your scanner cannot obtain enough slices in a single scan, then perform multiple scans with the specified TR/TE and with the maximum number of slices allowed by the system. Repeat the scans, each with the specified scan time, until all 11 transaxial images have been obtained in the proper locations.

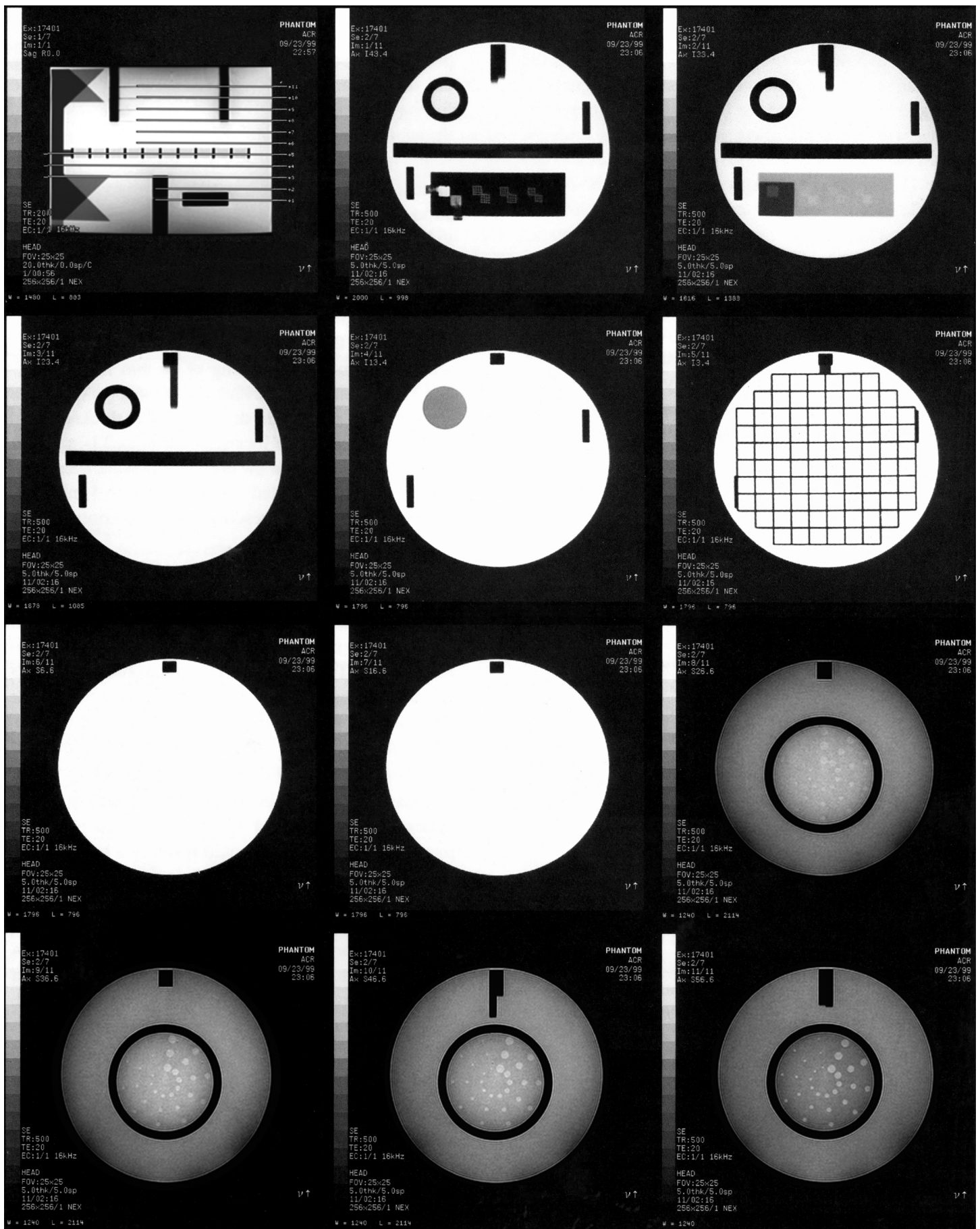


Figure 4: The sagittal localizer and all 11 slices of the SE 500/20 transaxial scan filmed in 12 on 1 format.

If your scanner is not capable of obtaining 5 mm slices with 5 mm gaps, then use the closest slice thickness to 5 mm for the specified TR and TE. Set the slice gap so that slice thickness plus slice gap equals 10 mm and be careful that the images are positioned as specified in Figure 2. If necessary, perform multiple scans, each with the specified scan parameters, to get these 11 images in as close to the proper locations as possible. Record each of these 11 transaxial images on the same sheet of film as the sagittal localizer image. The full sheet should look like the 11 images in Figure 4 in terms of window settings and image positions on the film. Please note: Your axial slices must be positioned as shown in Figure 2 in order for your images to be acceptable for evaluation. If your MRI system is unable to prescribe a 5-mm slice gap then you should try the following alternative slice positioning method: Either perform an interleaved multislice acquisition or perform 11 single slice acquisitions. Please make sure that each slice is positioned as shown in Figure 2. If you are unable to perform the axial slice positioning as indicated in this manual, then stop and contact the ACR before proceeding any further.

The conventional spin-echo (SE) 500/20 scan should be acquired with one acquisition per phase encoding step (one signal average acquisition or NEX) and the bandwidth used routinely for brain studies. This should take a total scan time of approximately two minutes. Enter the exact scan time required, along with the bandwidth (in kHz) on the Site Scanning Data Form. Remember to place check marks below the scan parameters that are used exactly as they appear on the table of Pulse Sequence Acquisition Parameters or enter the alternative parameters in each blank.

Acquire the SE 2000/20, 80 double-echo scan with one acquisition per phase encoding step at the same 11 slice locations as used for the previous scan. If a double echo at TEs of 20 and 80 ms cannot be obtained, then use the closest multiecho TEs to 20 and 80 ms (e.g., 40 and 80 ms). This scan should take approximately 8.5 minutes. (Record the sagittal localizer and each of the 11 SE 2000/80 images only on one sheet of film. The PD weighted images do not need to be filmed.) Enter the bandwidth for the scan in the space provided on the Site Scanning Data Form; enter the exact scan time required in the blank below scan time on the Data Form if it differs from the scan time specified on the form. Place check marks or enter the revised scan parameter in each block of the Data Form.

The ACR protocols do not specify any scan options, such as autoshim or image filtering, and you are not required

to use any. If you wish, you may use the scan options you normally use for clinical head imaging, provided those options do not interfere with attaining the scan parameters and slice prescriptions specified for the ACR protocols. Record all scan options used in the space provided on the Data Form.

For the ACR protocols, on scanners that have a range of image filter settings available, we recommend against strong filter settings because they are often detrimental to high-contrast resolution.

Next, scan the phantom using your site's T1- and T2-weighted scan protocols. It is important to acquire images with 5 mm slice thickness, if possible, or as close to 5 mm slice thickness as possible, and to acquire slices with center-to-center spacing of 10 mm for both T1- and T2-weighted images. Please try to adapt your normal scan protocols to obtain the 5 mm slice thickness and the specified 11 slice locations for both T1- and T2-weighted images. Enter the precise scan parameters used for T1- and T2-weighted scans (adapted to 5-mm slice thickness and the 11 prescribed slice locations) in the Site Scanning Data Form.

When adapting your site's sequences, only change slice thickness and slice spacing. Do not change other scan parameters. Many sites normally use a reduced field of view in the right-left dimension on their axial head images. This leads to wrap-around (aliasing) artifact when scanning the ACR phantom. Do not change the field of view of your sequences to avoid this artifact. This artifact does not interfere with our assessment of the images, and you will not be penalized for it.

Please film the sagittal localizer and each of the 11 slices for both T1- and T2-weighted scan protocols used by your site. This should result in a total of four sheets of film acquired in 12 on 1 format. Next, transfer the sagittal locator scan and the four sets of transaxial scans to a DICOM formatted CD-ROM. For more information on this, please refer to the instructions for digital data submission provided elsewhere in your accreditation package. Use a separate CD-ROM for each magnet. Do not store any clinical exams on the CD-ROM.

Submit data forms, films, and images on CD-ROM to the American College of Radiology.

If you have any suggestions or comments about the phantom image Site Scanning Instructions, please include those on the back of the Site Scanning Data Form.

MRI Phantom – Site Scanning Data Form

Please complete one copy of these data for each MR Magnet being evaluated. Photocopy this blank form for additional magnets. Detailed instructions for scanning the MRI phantom are attached. All information on this data sheet must be accurately specified. Please print or type. Please place your Phantom Data Form Label in the space above. **Return completed form with phantom images.**

1. MR Manufacturer: check one

- | | | | | |
|--------------------------------------|---|--------------------------------------|---------------------------------------|---|
| ☐ ELS Elscint | ☐ HI Health Images | ☐ OTS Otsuka | ☐ RES Resonex | ☐ TCH Technicare |
| ☐ FON Fonar | ☐ HIT Hitachi | ☐ PIC Picker | ☐ SIE Siemens | ☐ TOS Toshiba |
| ☐ GE GE | ☐ IN Instrumentarium | ☐ PHI Philips | ☐ SHI Shimadzu | ☐ OTH Other
specify _____ |

2. Model Name: _____

3. Serial Number: _____

4. Software Version: _____

5. Year Manufactured: _____

6. Magnetic Field Strength: check one

- | | | | |
|---|---|---|--|
| ☐ ¹ 0.064 | ☐ ² 0.2 T | ☐ ³ 0.3 T | ☐ ⁴ 0.35 T |
| ☐ ⁵ 0.5 T | ☐ ⁶ 1.0 T | ☐ ⁷ 1.5 T | ☐ ⁹⁹ Other specify |

7. Operating Location: check one

- | | | | |
|---|---|--|---|
| ☐ ⁶ Fixed | ☐ ² Fixed Trailer | ☐ ³ Mobile Trailer | ☐ ⁹ Other specify |
|---|---|--|---|

Pulse Sequence Acquisition Parameters

In the box below each parameter:
Record actual values if they differ from the prescribed protocol parameters or
Place a check mark to indicate use of prescribed parameter.
Fill in **all** parameters for "Your Site's Axial T1- and T2-weighted Brain Scan."

	a	b	c	d	e	f	g	h	i	j	k	l
	Study	Pulse Sequence	TR (ms)	TE (ms)	FOV (cm)	Number of Slices	Slice Thickness (mm)	Slice Gap (mm)	NEX	Matrix	Routine Receive Band-Width (kHz)	Scan Time (min:sec)
8.	ACR Sagittal locator	Spin Echo	200	20	25	1	20	N/A	1	256 256		0:56
								N/A				
9.	ACR Axial T1	Spin Echo	500	20	25	11	5	5	1	256 256		2:16
10.	ACR Axial T2 Double-echo	Spin Echo	2000	20/80	25	11	5	5	1	256 256		8:56
				/								
11.	Your Site's Axial T1 weighted Brain Scan				Freq:	11	5	5				
					Phase:							
12.	Your Site's Axial T2 weighted Brain Scan				Freq:	11	5	5				
					Phase:							

13. Scan Options Used on the ACR Spin-echo T1- and T2-weighted Axial Scans: _____

14. Scan Options Used on "Your Site's Axial T1- and T2-weighted Brain Scans:" _____

15. Serial number of phantom used for testing _____

Date of Testing: _____ Testing Performed by: _____ Phone: _____

(Please Print)

